CatNet: A Triangulated Evidence Framework for Income Classification

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Abstract

We present **CatNet** (Complementary Analysis and Triangulated Network), a reproducible six-stage framework for structured income classification analysis on the UCI Adult Census dataset (Dua and Graff, 2019). Rather than optimising a single performance metric, CatNet addresses four complementary research questions concerning distributional shape, feature dependency, sparse predictive signal, and decision boundary geometry through six coordinated modules: **WhiskerFlow** (preprocessing), **KittenGauss** (class-conditional distribution analysis), **TailMI** (mutual information feature analysis), **ClawReg** (L1-regularised logistic regression), **WhiskerInteract** (interaction validation), and **PawSVM** (kernel boundary analysis). MeowRank scoring identifies `education_num`, `age`, and `hours_per_week` as the strongest continuous discriminators. TailMI reveals strong structural dependencies—notably between `marital_status` and `relationship`—that do not translate into useful interaction terms under WhiskerInteract validation. ClawReg achieves a held-out ROC-AUC of 0.897 with a compact, stable feature core. Among the three PawSVM variants—**TabbySVM** (linear), **PurrPoly** (degree-2 polynomial), and **FurRBF** (radial basis function)—PurrPoly achieves the best ROC-AUC of 0.905, while FurRBF does not outperform TabbySVM, indicating modest low-order nonlinearity rather than broad kernel complexity. PurrRobust stability checks confirm all main conclusions across random seeds, regularisation levels, and MI discretisation schemes.

1 Introduction

Predicting whether an individual's annual income exceeds \$50,000 from census data has been a canonical binary classification benchmark since its introduction by Kohavi (1996). The UCI Adult dataset (Dua and Graff, 2019), derived from the 1994 United States Census, combines continuous demographic variables with categorical socioeconomic features and exhibits moderate class imbalance, making it a rich testbed for understanding the statistical structure of real-world classification problems. Most prior work on this dataset focuses on prediction performance—reporting accuracy or F1 improvements from model-level changes—without asking *why* any particular model performs as well as it does.

This paper takes a structural perspective. We treat the income prediction task as a data analysis problem, not merely a prediction benchmark, and organise the investigation around four complementary research questions: (1) What are the class-conditional distributional shapes of the continuous features, and which are well-described by a Gaussian summary? (2) How do features depend on one another at the level of mutual information, and do those dependencies suggest useful interaction terms? (3) Which features carry robust predictive signal under L1-enforced sparsity, and are the stable features consistent across the regularisation path? (4) Is the income decision boundary predominantly linear, or does it exhibit meaningful low-order nonlinear structure?

To answer these questions, we introduce **CatNet**, a Complementary Analysis and Triangulated Network framework consisting of six coordinated analytical modules. The central design principle of CatNet is triangulation: each module provides a distinct analytical lens, and the final interpretation is a synthesis of converging evidence rather than a conclusion from any single model. This design guards against the common risk of over-interpreting single-method findings—a feature can appear informative under one marginal view and yet become redundant in a sparse multivariate model, and a strong feature–feature dependency can reflect structural overlap rather than a productive interaction term.

The main contributions of this paper are as follows. We describe a reproducible six-stage analysis pipeline, CatNet, that systematically addresses structural questions about tabular classification data. We demonstrate through TailMI that the strongest feature dependencies in this dataset—including the `marital_status`–`relationship` pair—do not translate into useful interaction terms under explicit validation. We show through ClawReg that the strongest predictive signal is concentrated in a compact, stable set of feature groups. We establish through PawSVM that a degree-2 PurrPoly boundary captures modest but consistent nonlinear structure, while FurRBF does not improve over TabbySVM, pointing to limited low-order nonlinearity. Finally, PurrRobust stability checks confirm that all main conclusions hold across random seeds, regularisation choices, and MI discretisation schemes.

The remainder of the paper is organised as follows. Section 2 describes the system design. Section 3 covers the datasets used. Section 4 presents the experimental methodology. Section 5 presents results for each module. Section 6 synthesises the findings, and Section 7 concludes.

2 System Design

CatNet integrates six modules in a fixed sequential order. Each module produces structured outputs—figures, tables, and summary statistics—that feed into the next stage. All hyperparameter tuning is performed exclusively on the training split. The following subsections describe each module in detail.

2.1 WhiskerFlow: Data Preprocessing

WhiskerFlow transforms the raw UCI Adult records into a form suitable for downstream analysis. The input dataset contains 48,842 rows with a mix of continuous and categorical predictors. Two features are removed prior to analysis: `education`, because it duplicates the numeric `education_num`; and `fnlwgt`, following the project specification. The retained predictors are five continuous variables (`age`, `education_num`, `capital_gain`, `capital_loss`, `hours_per_week`) and seven categorical variables (`workclass`, `marital_status`, `occupation`, `relationship`, `race`, `gender`, `native_country`).

A key preprocessing step is the log-plus-one transform applied to `capital_gain` and `capital_loss`:

$$\tilde{x} = \log(1 + x), \quad x \geq 0. \tag{1}$$

This transform compresses the long right tails of these zero-inflated variables without discarding the zero mass, producing derived features `capital_gain_log1p` and `capital_loss_log1p` that are used in ClawReg. The `native_country` variable is collapsed into a binary grouped feature `native_country_grouped`, since approximately 89.7% of rows fall under the United States category and this variable carries very low feature–label MI (0.0006; see Section 5.3).

The main split is a stratified 80/20 train–test partition with random seed 42, yielding 39,073 training and 9,769 test rows. Stratification preserves the class ratio in both splits. Robustness reruns use seeds 7, 42, and 99 on the full dataset with the same protocol.

2.2 KittenGauss: Class-Conditional Distribution Analysis

KittenGauss fits univariate Gaussian distributions to each continuous feature conditional on the income class label. The class-conditional model is $p(x \mid y = c) = \mathcal{N}(\mu_c, \sigma_c^2)$, where parameters μ_c and σ_c are estimated from training-split data by maximum likelihood.

Feature discrimination is quantified using two complementary measures. The **MeowRank** score is the single-feature held-out ROC-AUC, computed by using the feature value alone as a classifier score. Cohen's d (Cohen, 1988) measures the standardised mean difference between classes:

$$d = \frac{\bar{x}_{>50K} - \bar{x}_{\leq 50K}}{s_{\text{pooled}}}, \tag{2}$$

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}.$$

KittenGauss is used as a descriptive lens rather than a generative claim: it is valuable for ranking features and identifying distributional anomalies, but it is not assumed that the data are actually Gaussian.

2.3 TailMI: Feature Dependency Analysis

TailMI quantifies pairwise statistical dependence using empirical mutual information estimates. For any pair of discrete or discretised variables X and Y, the empirical estimate is:

$$\hat{I}(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \hat{p}(x, y) \log_2 \frac{\hat{p}(x, y)}{\hat{p}(x)\,\hat{p}(y)}, \tag{3}$$

where joint and marginal distributions are estimated from training-split frequency counts. Continuous variables are discretised into equal-width bins before computing MI. Two binning schemes—a coarser and a finer one—provide a sensitivity check on the stability of the dependency ranking.

TailMI produces two outputs: a feature–label MI ranking, identifying which predictors carry the strongest marginal association with the income label; and a feature–feature MI matrix, identifying which pairs of predictors share the most redundant information. Both outputs feed into the WhiskerInteract candidate shortlist.

2.4 ClawReg: Sparse L1-Regularised Logistic Regression

ClawReg fits an L1-regularised logistic regression model (the **ClawLoss** objective) across a grid of regularisation strengths. ClawLoss is defined as:

$$\begin{aligned}\mathcal{L}_{\text{Claw}}(\boldsymbol{w}; C) = -\sum_{i=1}^{n} \Big[& y_i \log \sigma(\boldsymbol{w}^T \mathbf{x}_i) \\ & + (1 - y_i) \log\big(1 - \sigma(\boldsymbol{w}^T \mathbf{x}_i)\big) \Big] \\ & + \frac{1}{C} \|\boldsymbol{w}\|_1, \end{aligned} \tag{4}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the sigmoid function and $C > 0$ controls inverse regularisation strength. The regularisation grid spans nine logarithmically spaced values from $C = 0.001$ to $C = 10.0$. The best C is selected by 5-fold cross-validated ROC-AUC on the training split. Feature group stability is assessed by the fraction of grid points at which each group has at least one non-zero coefficient.

2.5 WhiskerInteract: Targeted Interaction Validation

WhiskerInteract tests a pre-specified set of numeric interaction candidates for meaningful improvement over the main-effects ClawReg model. Candidates are shortlisted from the TailMI feature–feature matrix and domain plausibility, restricted to numeric pairs to avoid the combinatorial explosion of categorical interactions. For each candidate, a feature equal to the product of the two standardised predictors is appended to the feature matrix and a new ClawLoss model is fitted. An interaction is considered potentially useful only if it yields a cross-validated ROC-AUC improvement exceeding a threshold of 0.001 over the main-effects baseline.

2.6 PawSVM: Kernel Boundary Shape Analysis

PawSVM compares three SVM variants to assess the shape of the income decision boundary. All variants solve the soft-margin SVM primal:

$$\begin{aligned} & \min_{\boldsymbol{w}, b, \boldsymbol{\xi}} \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C \sum_{i=1}^{n} \xi_i \\ & \text{s.t. } y_i\big(\boldsymbol{w}^T \phi(\mathbf{x}_i) + b\big) \geq 1 - \xi_i, \quad \xi_i \geq 0, \end{aligned} \tag{5}$$

with different kernel functions $k(\mathbf{x}_i, \mathbf{x}_j)$. **TabbySVM** uses a linear kernel. **PurrPoly** uses a degree-d polynomial kernel:

$$k_{\text{PurrPoly}}(\mathbf{x}_i, \mathbf{x}_j) = \big(\gamma\, \mathbf{x}_i^T \mathbf{x}_j + r\big)^d. \tag{6}$$

FurRBF uses a radial basis function kernel:

$$k_{\text{FurRBF}}(\mathbf{x}_i, \mathbf{x}_j) = \exp\big(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\big). \tag{7}$$

Each variant is independently tuned by grid search on the training split using 5-fold cross-validated ROC-AUC. Kernel selection is made after tuning, based on held-out test ROC-AUC.

3 Datasets Used

The UCI Adult dataset was introduced as a machine learning benchmark by Kohavi (1996) and has since become one of the most frequently cited datasets for evaluating classifier performance, fairness interventions, and interpretability methods. Standard preprocessing choices—removing the `fnlwgt` sample weight and the string-valued `education` in favour of the numeric `education_num`—have become conventional across the literature. Despite the dataset's ubiquity, most published studies treat it as a black-box benchmark rather than as a subject of structural investigation.

Mutual information provides a principled, non-parametric measure of statistical dependence (Shannon, 1948; Cover and Thomas, 2006). Unlike linear correlation, it captures arbitrary functional dependencies, making it well-suited for identifying nonlinear feature–label relationships and characterising redundancy among feature pairs. Filter-based feature selection methods based on MI have been studied extensively (Hastie et al., 2009); here, TailMI uses MI primarily as a diagnostic and dependency characterisation tool rather than as a selection filter.

L1-regularised logistic regression, popularised as the LASSO by Tibshirani (1996), enforces exact coefficient sparsity and has become a standard tool for identifying compact sets of informative features. Andrew and Gao (2007) demonstrate efficient training algorithms for L1-regularised log-linear models.

Our ClawReg module studies the full regularisation path rather than a single operating point, allowing us to assess which feature groups consistently survive as sparsity increases and thereby distinguish robust signals from ones that disappear under mild regularisation.

Support vector machines (Cortes and Vapnik, 1995), extended to nonlinear boundaries through the kernel trick (Boser et al., 1992), provide a principled framework for comparing boundary shapes. The choice among linear, polynomial, and radial basis function kernels directly encodes assumptions about the complexity of the decision surface. ROC-AUC is used as the primary evaluation metric throughout this paper because it is insensitive to the class imbalance present in this dataset and provides a consistent ranking across probability thresholds (Fawcett, 2006; Hand, 2009). All implementations use scikit-learn (Pedregosa et al., 2011).

4 Methodology

All experiments use the WhiskerFlow 80/20 stratified train–test split with seed 42 as the primary evaluation setting. Hyperparameter tuning is performed exclusively on the training split using 5-fold stratified cross-validation. The test split is reserved for final evaluation and robustness reruns, and is never used for model selection.

Categorical features are one-hot encoded before being fed into ClawReg and PawSVM. Continuous features are standardised to zero mean and unit variance on the training split statistics. The TabbySVM grid uses $C \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$. PurrPoly uses $d \in \{2, 3\}$, $C \in \{0.1, 1.0, 10.0\}$, and $r \in \{0.0, 1.0\}$. FurRBF uses $C \in \{1.0, 3.0, 10.0\}$ and $\gamma \in \{0.01, 0.05, 0.1, 0.5\}$. Evaluation uses accuracy, macro-averaged precision and recall, positive-class F1, and ROC-AUC, with ROC-AUC as the primary metric throughout. All implementations use scikit-learn 1.8.0 (Pedregosa et al., 2011) with Python 3.12.

5 Results

5.1 Focused Exploratory Analysis

The training split contains 39,073 rows, of which 29,724 (76.1%) belong to the `<=50K` class and 9,349 (23.9%) belong to the `>50K` class. This imbalance, illustrated in Figure 1, motivates the use of ROC-AUC and F1 as primary metrics.

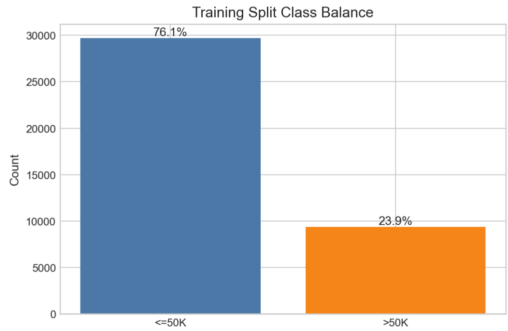

Figure 1: Class distribution in the training split. The `>50K` class constitutes 23.9% of training instances, motivating ROC-AUC as the primary evaluation metric.

Three notable distributional patterns emerge from the continuous feature flags: `capital_gain` is extremely sparse (91.8% zeros), `capital_loss` is even sparser (95.2% zeros), and `hours_per_week` has a dominant repeated value (46.5% of rows at a single value). These patterns motivate the WhiskerFlow log1p transforms and inform the interpretation of KittenGauss results. Figure 2 shows the class-conditional distribution of all five continuous features, with the visible separation in `education_num` and `age` contrasting with the zero-dominated distributions of the capital variables.

Preliminary inspection of categorical variables reveals strong differences in positive-income rates across groups. Figure 3 summarises the most striking contrasts: `Married-civ-spouse` has a 44.8% positive rate versus 4.4% for `Never-married`; `Exec-managerial` has 48.2% versus 4.0% for `Other-service`; and `Male` has 30.4% versus 10.9% for `Female`. These large group differences already suggest that household structure and occupation-related variables will dominate the multivariate models.

5.2 KittenGauss Distribution Analysis

Table 1 summarises the class-conditional Gaussian statistics and MeowRank scores for the five continuous features. `education_num` is clearly the strongest continuous discriminator, with a MeowRank ROC-AUC of 0.716 and a Cohen's d of 0.825. `age` and `hours_per_week` show moderate but meaningful separation, while the capital variables have identical class medians of zero, reflecting their extreme zero inflation.

Figure 4 shows the empirical histograms overlaid with the fitted class-conditional Gaussian curves. The approximation is reasonable for `education_num`, `age`, and `hours_per_week`,

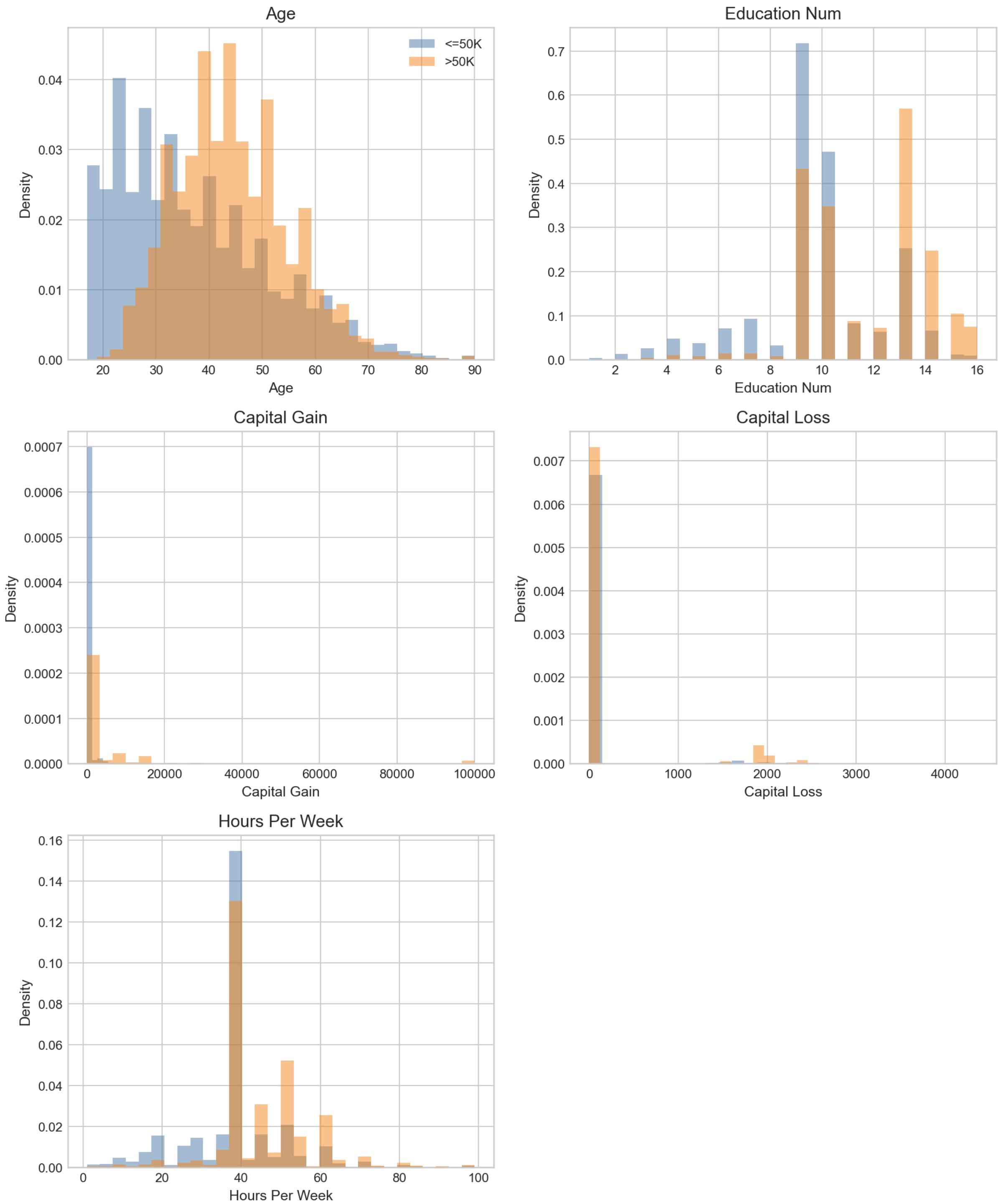

Figure 2: Continuous feature distributions by income class. `education_num`, `age`, and `hours_per_week` show clear class separation. `capital_gain` and `capital_loss` are dominated by the zero mass and long right tails, making the Gaussian summary a poor description of their shape.

but breaks down for `capital_gain` and `capital_loss`, whose distributions consist of a large zero mass and a sparse right tail that no Gaussian can accommodate. Figure 5 ranks all five features by MeowRank score, confirming that `education_num` is the dominant marginal separator.

5.3 TailMI Feature Dependencies

Table 2 reports the top feature–label and feature–feature MI values. The leading feature–label MI values belong to `relationship` (0.116) and `marital_status` (0.111), substantially above the

Feature	**Mean** ($\leq$50K)	**Mean** (>50K)	**Cohen's** d	**MeowRank AUC**
`education_num`	9.60	11.60	0.825	0.716
`age`	36.92	44.35	0.556	0.682
`hours_per_week`	38.89	45.43	0.541	0.671
`capital_gain`	147.7	3950	0.532	0.588
`capital_loss`	55.3	200.0	0.358	0.535

Table 1: KittenGauss statistics for the five continuous features. MeowRank AUC is the single-feature held-out ROC-AUC. Despite the large mean differences for the capital variables, their MeowRank scores are low because the Gaussian summary is a poor fit for zero-inflated distributions.

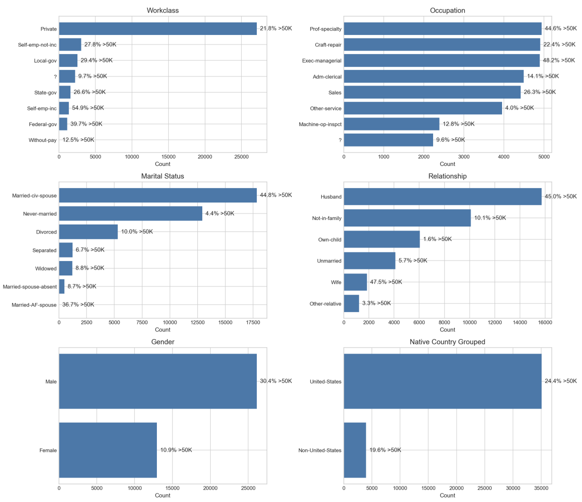

Figure 3: Selected categorical feature distributions with positive-income rates. Household structure (`marital_status`, `relationship`) and occupation-related variables already show large income-class differences before formal modelling.

numeric features (age: 0.064, `education_num`: 0.062). The strongest pairwise dependencies are concentrated among household-structure and work-context variables: `marital_status`–`relationship` ($\hat{I} = 0.725$), `workclass`–`occupation` (0.329), and `relationship`–`gender` (0.270).

The full TailMI pairwise MI matrix is displayed in Figure 6. Figures 7 and 8 show the feature–label MI ranking and the top feature–feature pairs as bar charts. The discretisation sensitivity check confirms complete overlap in the top-10 MI pairs (10/10) and in the candidate interaction shortlist (5/5) across the two binning schemes, indicating that the TailMI dependency picture is stable with respect to discretisation choice.

5.4 ClawReg Sparse Model

The best ClawReg model selects $C = 1.0$ by cross-validation and achieves the held-out performance shown in Table 3. The ROC-AUC of 0.897 is the

Feature	**Feature–label MI**
`relationship`	0.116
`marital_status`	0.111
`age`	0.064
`occupation`	0.063
`education_num`	0.062
`capital_gain`	0.054

Feature pair	**Pairwise MI**
`marital_status` × `relationship`	0.725
`workclass` × `occupation`	0.329
`relationship` × `gender`	0.270
`age` × `marital_status`	0.233
`education_num` × `occupation`	0.213

Table 2: TailMI results. *Left*: top feature–label MI values. *Right*: top five feature–feature MI pairs. The high pairwise MI reflects structural overlap among sociodemographic variables rather than indicating useful interaction candidates.

primary result.

Acc.	**Prec.**	**Rec.**	**F1**	**AUC**	**Best** C
0.848	0.723	0.589	0.649	0.897	1.0

Table 3: ClawReg held-out performance on the test split (seed 42). Acc. = accuracy, Prec. = precision, Rec. = recall, AUC = ROC-AUC.

The cross-validated ROC-AUC rises from 0.887 at $C = 0.001$ to a plateau near 0.900 by $C = 0.1$. The plateau is very flat: the range from $C = 0.1$ to $C = 10.0$ spans only 0.0003 in mean CV ROC-AUC (0.8998, 0.9001, 0.9001, 0.9001 at $C = 0.1, 0.316, 1.0, 10.0$ respectively). The logistic model's performance is therefore largely insensitive to the exact regularisation level once moderate sparsity is allowed.

Table 4 reports the feature groups that remain non-zero across the full ClawLoss regularisation grid. Six groups are perfectly stable (non-zero at all 9 grid points): `marital_status` dominates with a mean absolute coefficient of 3.359, followed by `education_num` (0.741), `capital_gain_log1p`

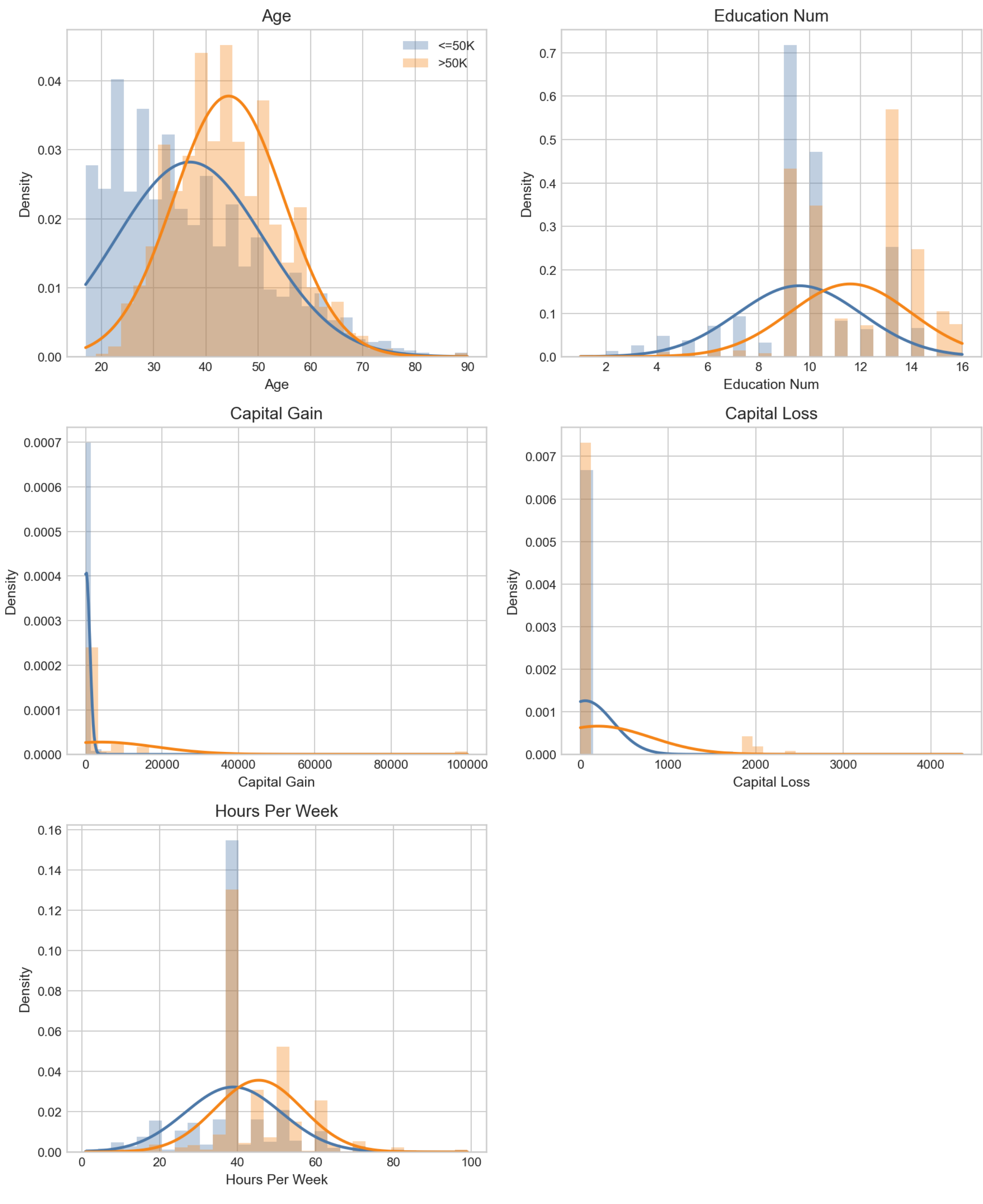

Figure 4: KittenGauss fitted overlays. Empirical histograms with class-conditional Gaussian curves for each continuous feature. The Gaussian fit is adequate for `education_num`, age, and `hours_per_week`, but fails completely for the zero-inflated capital variables.

(0.475), `hours_per_week` (0.328), `age` (0.317), and `capital_loss_log1p` (0.220).

Figure 9 shows the ClawLoss coefficient paths across the regularisation grid. The contrast between the core stable groups—especially `marital_status`, whose coefficient remains large throughout—and the peripheral groups that enter and leave the active set as C increases illustrates the concentration of predictive signal.

5.5 WhiskerInteract Interaction Validation

WhiskerInteract tests five numeric interaction candidates drawn from the TailMI shortlist. Table 5

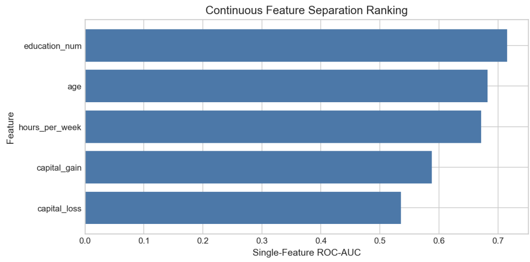

Figure 5: MeowRank: continuous feature ranking by single-feature held-out ROC-AUC. `education_num` is clearly the strongest marginal continuous discriminator.

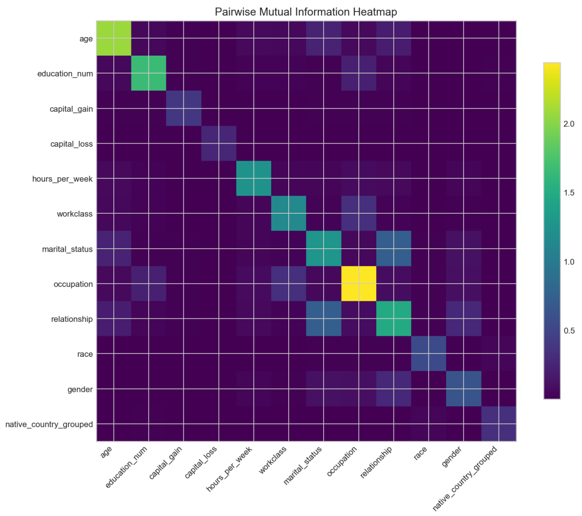

Figure 6: TailMI pairwise mutual information heatmap. The strongest dependencies are concentrated among household-structure and work-context variables, not in the numeric interaction shortlist. Darker cells indicate higher MI values.

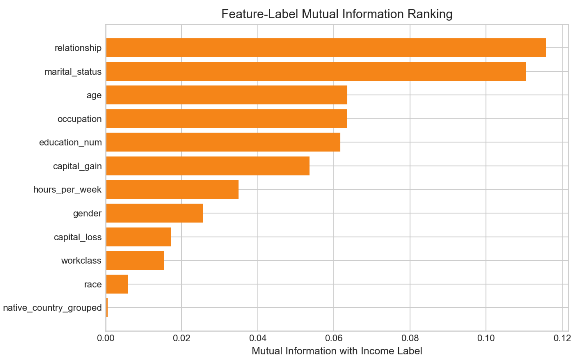

Figure 7: TailMI feature–label MI ranking. `relationship` and `marital_status` lead, with numeric features further down the ranking.

summarises the results. Every tested interaction produces changes in cross-validated ROC-AUC smaller than 10^{-3} in magnitude. The largest positive change in CV ROC-AUC is only $+0.000174$ for `age` $\times$ `hours_per_week`, and no interaction

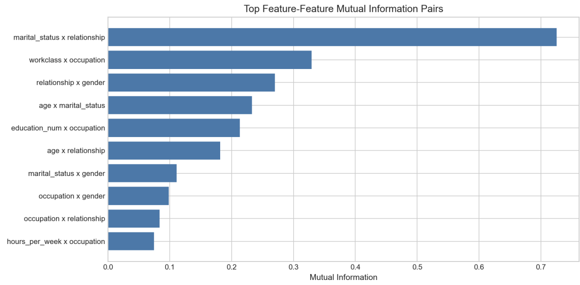

Figure 8: Top feature–feature MI pairs. The dominant pairs reflect sociodemographic overlap rather than productive interaction candidates.

Feature group	**Non-zero share**	**Mean $\lvert$coef$\rvert$**
`marital_status`	1.000	3.359
`education_num`	1.000	0.741
`capital_gain_log1p`	1.000	0.475
`hours_per_week`	1.000	0.328
`age`	1.000	0.317
`capital_loss_log1p`	1.000	0.220
`occupation`	0.889	—
`relationship`	0.778	—
`workclass`	0.778	—
`gender`	0.778	—

Table 4: ClawReg feature group stability across the full regularisation grid. Non-zero share is the fraction of the 9 grid points at which the group has at least one non-zero coefficient. Groups with non-zero share < 1 do not appear at all grid points.

consistently improves both the cross-validation and test metrics simultaneously.

Because no interaction clears the 0.001 threshold and the test-set delta signs are inconsistent across candidates, no interaction-augmented final model is retained. Figure 10 summarises the delta values visually.

5.6 PawSVM Boundary Analysis

Table 6 compares the three PawSVM variants. TabbySVM achieves a test ROC-AUC of 0.897 at $C = 0.1$. PurrPoly with degree 2 achieves 0.905, an improvement of approximately 0.008. FurRBF achieves 0.896, fractionally below TabbySVM.

The degree-3 PurrPoly variant is clearly inferior in cross-validation (CV ROC-AUC ≈ 0.883) compared to degree-2 (0.898), indicating that useful nonlinearity is low-order. The FurRBF result is consistent with the WhiskerInteract finding: the problem does not require flexible local boundaries, and the PurrPoly improvement is more plausibly due to capturing low-order polynomial effects across the encoded feature space. Figure 11 summarises the held-out ROC-AUC comparison.

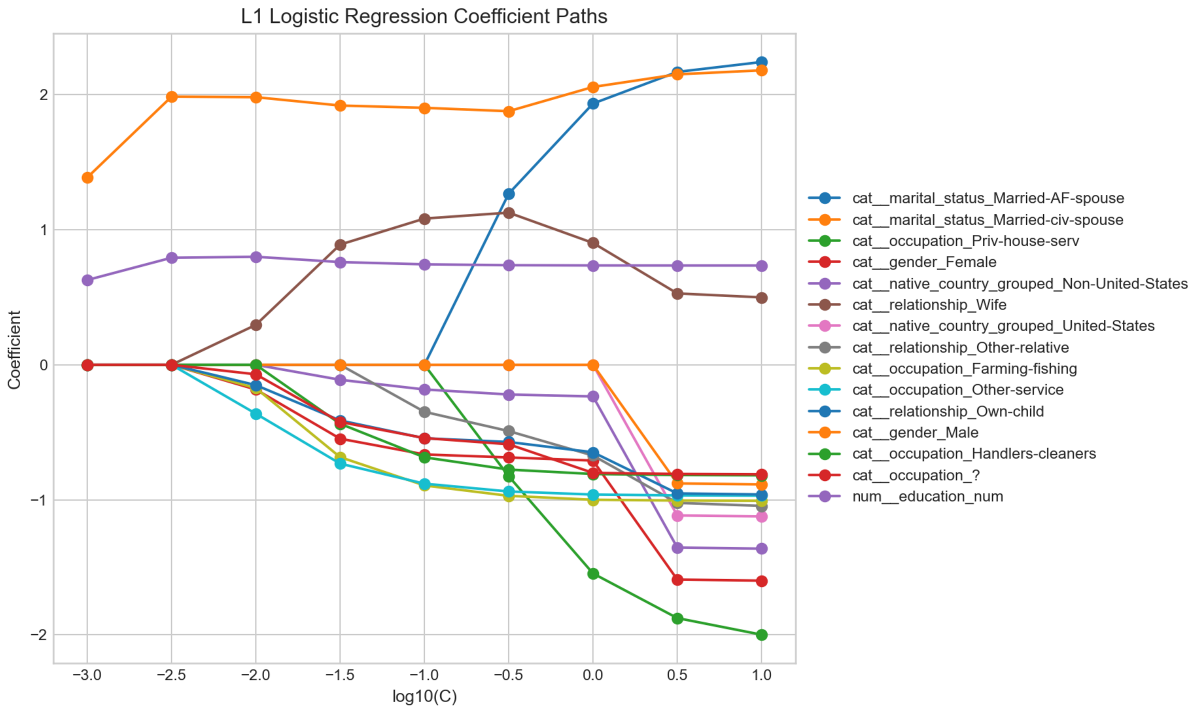

Figure 9: ClawReg coefficient paths across the L1 regularisation grid (ClawLoss). A compact set of feature groups remains active across the full range of regularisation strengths. The scale of the `marital_status` coefficient reflects both its predictive strength and the one-hot encoding magnitudes of individual categories.

Interaction	**Pairwise MI**	**Δ CV AUC**	**Δ Test AUC**
`age` × `hours_per_week`	0.066	+0.000174	+0.000210
`age` × `education_num`	0.052	−0.000026	−0.000012
`education_num` × `hours_per_week`	0.027	−0.000043	−0.000003
`education_num` × `capital_gain`	0.016	−0.000025	+0.000057
`capital_gain` × `hours_per_week`	0.008	+0.000042	−0.000066

Table 5: WhiskerInteract results. All ROC-AUC deltas fall below the 0.001 improvement threshold. No interaction-augmented model was retained.

Model	**Best hyperparameters**	**CV AUC**	**Test AUC**	**Test F1**
TabbySVM (linear)	$C = 0.1$	0.893	0.897	0.644
PurrPoly (poly-2)	$d = 2,\ C = 1.0,\ r = 1.0$	0.898	0.905	0.670
FurRBF (RBF)	$C = 3.0,\ \gamma = 0.05$	0.892	0.896	0.670

Table 6: PawSVM comparison across TabbySVM, PurrPoly, and FurRBF. PurrPoly achieves the highest held-out ROC-AUC. FurRBF does not improve over TabbySVM, ruling out the need for a highly flexible local boundary.

5.7 PurrRobust Stability Checks

Table 7 reports the held-out test ROC-AUC for the three main models across random seeds 7, 42, and 99. The range across seeds is narrow for all models: ClawReg spans 0.897–0.901, TabbySVM spans 0.897–0.901, and PurrPoly spans 0.905–0.906. PurrPoly remains above TabbySVM at every seed, with margins of +0.0063, +0.0078, and +0.0045 at seeds 7, 42, and 99 respectively.

Model	**Seed 7**	**Seed 42**	**Seed 99**	**Range**
ClawReg	0.901	0.897	0.901	0.004
TabbySVM	0.900	0.897	0.901	0.004
PurrPoly	0.906	0.905	0.906	0.001

Table 7: PurrRobust seed stability. Held-out test ROC-AUC across three stratified random splits. PurrPoly consistently leads at every seed. All models show narrow variation, confirming that the main conclusions do not depend on a specific random split.

The TailMI discretisation sensitivity check con-

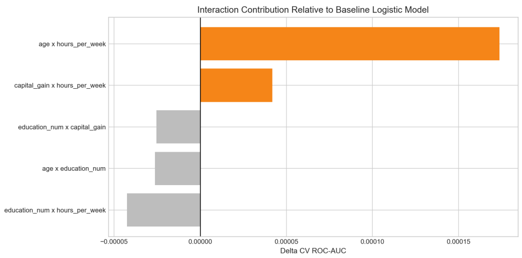

Figure 10: WhiskerInteract: incremental change in CV ROC-AUC from adding each interaction term. All effects are well below the 0.001 threshold, and no consistent pattern emerges.

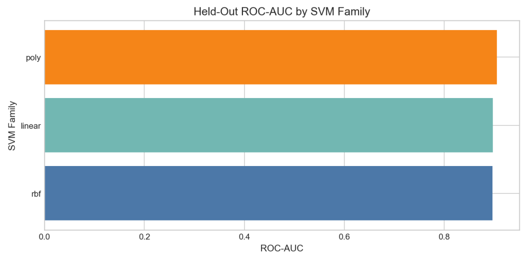

Figure 11: PawSVM held-out ROC-AUC comparison. PurrPoly (degree-2 polynomial) consistently outperforms TabbySVM (linear) and FurRBF (RBF), confirming modest low-order nonlinearity.

firms stability: both the top-10 MI pairs (10/10) and the candidate interaction shortlist (5/5) overlap completely across the two binning schemes. Figure 12 shows the metric ranges across seeds.

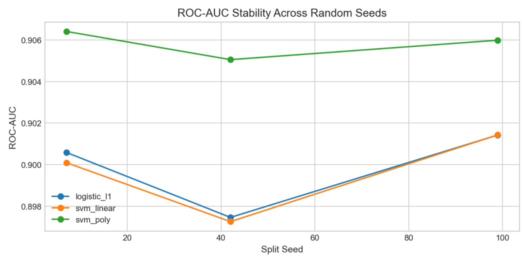

Figure 12: PurrRobust: ROC-AUC metric ranges across seeds 7, 42, and 99. The ordering of PurrPoly above TabbySVM and ClawReg is consistent across all three splits.

6 Evaluation Metrics Used

Taken together, the six CatNet modules paint a consistent picture of the Adult income dataset. The clearest signals are held by household-structure and education-related variables, a pattern independently confirmed by TailMI feature–label rankings, KittenGauss MeowRank scores, and the ClawReg feature stability analysis. This cross-method convergence is meaningful: the strong features are genuinely informative rather than artefacts of any particular modelling assumption.

A more nuanced finding emerges from the comparison between TailMI and ClawReg. `relationship` has the highest feature–label MI (0.116) but is less stable in ClawReg than `marital_status`, which has MI of 0.111. The most likely explanation is that these variables carry heavily overlapping information about household role, and once `marital_status` enters the model, `relationship` adds limited incremental signal. This redundancy is already visible in the large pairwise MI (0.725). The broader lesson is that MI rankings should not be read directly as a ranking of predictive utility in a multivariate model: high marginal relevance does not imply low redundancy.

The capital variables provide a useful case study in the value of transformation and triangulation. Under the KittenGauss lens, `capital_gain` and `capital_loss` are flagged as non-Gaussian and weak, because the Gaussian summary cannot accommodate zero-inflated distributions. After the WhiskerFlow log1p transform, however, `capital_gain_log1p` and `capital_loss_log1p` both survive the full ClawReg regularisation path. A finding that appears weak under one analytical lens should therefore not be dismissed without examining alternative representations.

The PawSVM boundary analysis resolves the fourth research question: the income decision boundary is not purely linear, but the nonlinearity is modest and low-order. The degree-2 PurrPoly kernel captures a consistent improvement of approximately 0.008 in ROC-AUC over TabbySVM across all three seeds, while FurRBF does not improve over the linear baseline. This suggests that the boundary is better described by a degree-2 polynomial surface than by a highly flexible radial kernel—consistent with the WhiskerInteract finding that explicit low-order numeric products provide negligible lift, since PurrPoly may be capturing a broader range of such effects across the full one-hot encoded feature space rather than the specific five products that WhiskerInteract tested.

7 Final Conclusions

We have introduced CatNet, a six-module triangulated evidence framework for structured income classification analysis on the UCI Adult dataset. CatNet integrates WhiskerFlow preprocessing, KittenGauss class-conditional distribution analysis, TailMI feature dependency analysis, ClawReg sparse logistic regression, WhiskerInteract interaction validation, and PawSVM kernel boundary comparison into a coherent analytical pipeline that addresses four research questions about distributional shape, feature dependencies, sparse predictive signal, and decision boundary geometry.

The main findings are that a compact core of features (`marital_status`, `education_num`, `capital_gain_log1p`, `hours_per_week`, `age`, and `capital_loss_log1p`) dominates the ClawReg sparse model; that strong TailMI dependencies among household-structure variables reflect redundancy rather than useful interactions; that log-transformed capital variables carry significant signal despite appearing weak under the Gaussian lens; and that the income boundary exhibits modest low-order polynomial nonlinearity captured by degree-2 PurrPoly but not by FurRBF. PurrRobust stability checks confirm all findings across three random seeds, nine regularisation levels, and two MI discretisation schemes.

Limitations. KittenGauss provides useful descriptive comparisons but is a poor model for zero-inflated variables such as `capital_gain`; findings from this module should be interpreted alongside the ClawReg results rather than in isolation. The WhiskerInteract validation covers only five numeric interaction products and does not explore categorical or mixed-type interactions, which could in principle yield larger improvements. The PawSVM comparison reveals the existence of modest nonlinearity but does not identify which specific polynomial terms drive the PurrPoly improvement. The missing-value handling in the original pipeline preserves the literal ? string as an explicit categorical level rather than imputing or removing affected rows, which may affect the categorical frequency summaries. Finally, all conclusions are specific to the UCI Adult dataset and the 1994 United States census population it represents; they should not be generalised to other income prediction settings without independent validation.

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